MediXO EduX
AI Academic Intelligence

EXAMINATION
INTELLIGENCE
REPORT

FULL EXAM ANALYSIS REPORT

DETAILED PERFORMANCE • QUESTION ANALYSIS • LEARNING INSIGHTS

JEE Main Figure Validation Paper

Combined (Physics, Chemistry, Mathematics) | All Subjects



56.9%
Score

Marks Obtained

41 / 72

8 Correct | 7 Incorrect | 3 Skipped

Student	**Aarav Sharma**
Roll Number	**21CS114**
Exam Date	**01 Aug 2026**
Total Marks	**72**

Subject Analysis | Chapter Mastery | Question-wise Review | Error & Learning Insights | Revision Plan & Next Steps

DISCIPLINE
CREATIVITY
PROGRESS

ABRIGHTER
TOMORROW
TOGETHER

01 PERFORMANCE INTELLIGENCE

Performance intelligence

Response composition

18 questions in the paper



- Correct 9
- Incorrect 7
- Skipped 2

Difficulty analysis

Questions and published accuracy

Easy		5 Q · 80%
Medium		8 Q · 50%
Hard		5 Q · 33%

Accuracy values are reproduced from the payload.

Mistake distribution

Count by published classification

Concept Error		3
Formula Error		2
Calculation Error		2

Source classifications; rows are measured independently.

Subject-wise performance

Accuracy by subject

Physics		55.6%
Chemistry		50%

Physics 19/36 Chemistry 14/24 Mathematics 8/12

Subject-wise performance

Subject-wise performance · continued

Accuracy by subject

Mathematics 40%

Physics 19/36 Chemistry 14/24 Mathematics 8/12

Important performance signals

GRADE **B**	RANK **2480**	PERCENTILE **87.4**
BATCH RANK **12**	COHORT SIZE **180**	OVERALL SPEED **3.2**
CONFIDENCE INDEX **71**	READINESS SCORE **66**	HEALTH SCORE **74**
NEGATIVE MARKS **7**	GUESS ATTEMPTS **1**	ATTEMPT RATIO **83.3%**
TOTAL QUESTIONS **18**	OVERALL ACCURACY **56.25%**	BADGE **Figure QA**

02 MASTERY MAP

Subject and chapter mastery

Subject marks, accuracy and classification

SUBJECT	MARKS	ACCURACY	RANK	LEVEL	STRONG AREAS	WEAK AREAS
Physics	19 / 36	55.6%	2	Medium	Circuits	Ray optics
Chemistry	14 / 24	50%	4	Hard	Thermodynamics	Bonding
Mathematics	8 / 12	40%	6	Hard	Vectors	Probability

Strong, weak and critical chapters by subject

SUBJECT	STRONG CHAPTERS	WEAK CHAPTERS	CRITICAL CHAPTERS
Physics	Rotational Motion	Electrostatics	
Chemistry		Chemical Bonding · Organic Chemistry	Chemical Bonding
Mathematics	Calculus	Probability	Probability

Chapter mastery · grouped by subject

CHAPTER	ACCURACY	MARKS	TIME	MASTERY
Physics				
Current Electricity	74%	8	9m	Average
Electrostatics	41%	4	11m	Weak
Rotational Motion	100%	4	6m	Strong
Chemistry				
Chemical Bonding	38%	4	8m	Critical
Organic Chemistry	44%	4	10m	Weak
Mathematics				
Calculus	100%	4	4m	Strong
Probability	32%	4	7m	Critical

03 PACE INTELLIGENCE

Time intelligence

Chapter time consumption · measured rows

CHAPTER	SHARE OF TOTAL TIME	TIME CONSUMED	SHARE
Physics			
Current Electricity	16%	9m	16%
Electrostatics	20%	11m	20%
Rotational Motion	11%	6m	11%
Chemistry			
Chemical Bonding	15%	8m	15%
Organic Chemistry	18%	10m	18%
Mathematics			
Calculus	7%	4m	7%
Probability	13%	7m	13%

Time-management metrics

AVERAGE PER QUESTION
3.2

NAVIGATION COUNT
4

MANAGEMENT SCORE
62

FASTEST QUESTION
Q3

FASTEST TIME
1.1

SLOWEST QUESTION
Q11

SLOWEST TIME
6.1

Sectional allocation and efficiency

SECTION	ALLOCATED	USED	EFFICIENCY
Physics	180	38	21
Chemistry	180	26	14
Mathematics	180	30	17

04 COMPARISON & TRAJECTORY

Comparison and projection

Comparative performance

Comparison register

BENCHMARK	SCORE	PERCENTILE	GRADE	REFERENCE
Previous test	38	84.2	B	
Batch average	39	62.1		Batch A

VS PREVIOUS TEST
+3

Projection and readiness

Score trajectory



Expected improvement +4.0% · Risk Medium

JEE PERCENTILE
92.4

EXPECTED AIR
9800

TARGET PROBABILITY
64

RISK LEVEL
Medium

EXPECTED IMPROVEMENT
+4.0%

Trajectory values

EXAMINATION / PROJECTION	SCORE
Mock 1	34
Figure validation	41

READINESS SCORE
66

CONFIDENCE INDEX
71

HEALTH SCORE
74

OVERALL ACCURACY
56.25%

BADGE
Figure QA

05 QUESTION INTELLIGENCE

Question-by-question analysis

18 questions reviewed · published explanations reproduced in full

01 Physics

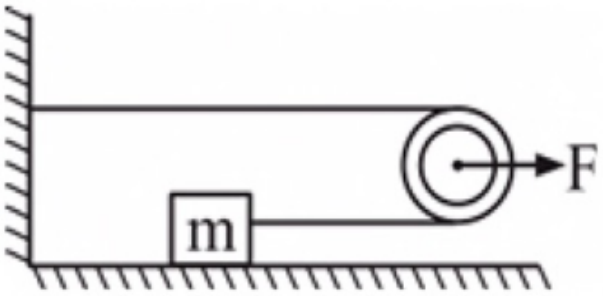
9 questions

QUESTION 01 PHYSICS

CORRECT

Current Electricity · Circuit analysis · Medium · MCQ · 4 marks · 2m 24s

In the circuit shown, the equivalent resistance between A and B is:



OPTIONS

A. The potential difference across the resistor is 6 V.

B. The current through the branch is 2 A.

C. The power dissipated in the circuit is 24 W.

D. A very long alternative answer that must wrap cleanly without clipping the question block or the option column.

STUDENT RESPONSE
B

CORRECT ANSWER
B

PERFORMANCE INSIGHT
Answered correctly — 4 marks

EXPLANATION

Complete solution. Apply the governing relation and substitute the values stated in the question.

$$\oint \mathbf{E} \cdot d\mathbf{A} = \frac{q_{enc}}{\varepsilon_0}$$

The closed surface encloses the charge distribution shown in the figure, so the normal component of the field is constant across the curved surface and vanishes across the flat ends. Evaluating the flux integral gives the result reported in the key. Check every sign

Physics · Q1 · explanation continued

convention, unit and limiting case before selecting the final option.

QUESTION 02 · PHYSICS

Electrostatics · Gauss law · Hard · MCQ · −1 mark · 4m 48s

Two statements are made about the field lines in the figure. Choose the correct option.

OPTIONS

A. Only statement I is correct.

B. Only statement II is correct.

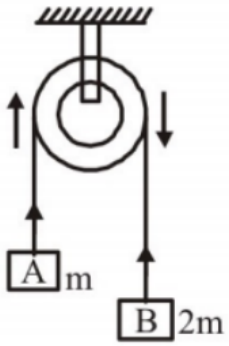
C. Both statements are correct.

D. Neither statement is correct.

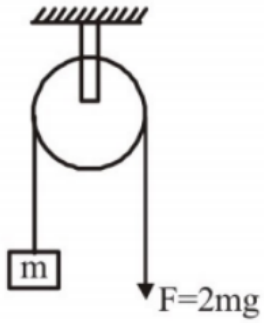
Student response: C

Correct answer: A

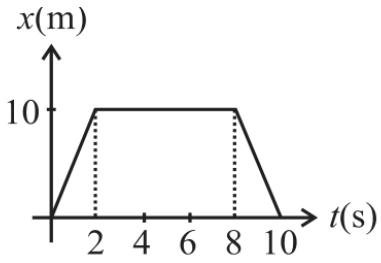
Answered incorrectly — 1 mark deducted · Concept Error · The field inside a conductor is zero in electrostatic equilibrium.



(a)

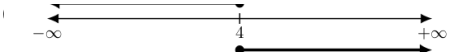


(b)

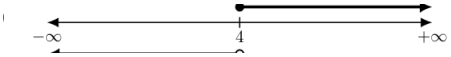


OPTION FIGURES

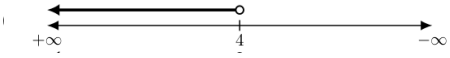
A.



B.



C.



D.



Physics · Q2 · explanation continued

EXPLANATION

Compare the flux through the two surfaces shown. Complete solution. Apply the governing relation and substitute the values stated in the question.

$$\oint \mathbf{E} \cdot d\mathbf{A} = \frac{q_{enc}}{\varepsilon_0}$$

The closed surface encloses the charge distribution shown in the figure, so the normal component of the field is constant across the curved surface and vanishes across the flat ends. Evaluating the flux integral gives the result reported in the key. Check every sign convention, unit and limiting case before selecting the final option.

MISTAKE NOTE

The field inside a conductor is zero in electrostatic equilibrium.

Physics · Q3 · question continued

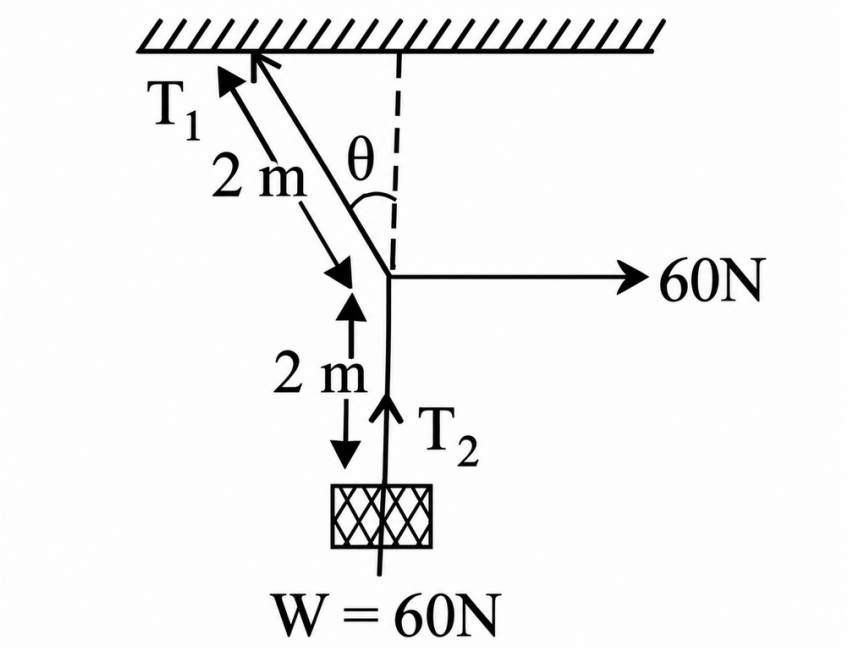
QUESTION 03 PHYSICS

SKIPPED

Rotational Motion · Moment of inertia · Easy · MCQ · 0 marks · 1m 06s

The moment of inertia of the rigid body shown about the axis is:

represents:



OPTIONS

A. The potential difference across the resistor is 6 V.

B. The current through the branch is 2 A.

C. The power dissipated in the circuit is 24 W.

D. A very long alternative answer that must wrap cleanly without clipping the question block or the option column.

STUDENT RESPONSE
Not attempted

CORRECT ANSWER
D

PERFORMANCE INSIGHT
Not attempted — no marks awarded

EXPLANATION

Use the parallel-axis theorem. Complete solution. Apply the governing relation and substitute the values stated in the question.

$$\oint \mathbf{E} \cdot d\mathbf{A} = \frac{q_{enc}}{\varepsilon_0}$$

The closed surface encloses the charge distribution shown in the figure, so the normal

Physics · Q3 · explanation continued

component of the field is constant across the curved surface and vanishes across the flat ends. Evaluating the flux integral gives the result reported in the key. Check every sign convention, unit and limiting case before selecting the final option.

QUESTION 04 PHYSICS

CORRECT

Modern Physics · Photoelectric effect · Medium · MCQ · 4 marks · 5m 12s

The graph in the figure plots stopping potential against frequency. Identify the correct interpretation.

OPTIONS

A. The potential difference across the resistor is 6 V.

B. The current through the branch is 2 A.

C. The power dissipated in the circuit is 24 W.

D. A very long alternative answer that must wrap cleanly without clipping the question block or the option column.

STUDENT RESPONSE
A

CORRECT ANSWER
A

PERFORMANCE INSIGHT
Answered correctly — 4 marks

EXPLANATION

The slope of the graph is h/e. Complete solution. Apply the governing relation and substitute the values stated in the question.

$$\oint \mathbf{E} \cdot d\mathbf{A} = \frac{q_{enc}}{\varepsilon_0}$$

The closed surface encloses the charge distribution shown in the figure, so the normal component of the field is constant across the curved surface and vanishes across the flat ends. Evaluating the flux integral gives the result reported in the key. Check every sign convention, unit and limiting case before selecting the final option.

Physics · Q5 · question continued

QUESTION 05 PHYSICS

INCORRECT

Optics · Ray optics · Hard · MCQ · −1 mark · 3m 18s

The ray diagram in the figure shows a lens combination. The final image is:

OPTIONS

A. The potential difference across the resistor is 6 V.

B. The current through the branch is 2 A.

C. The power dissipated in the circuit is 24 W.

D. A very long alternative answer that must wrap cleanly without clipping the question block or the option column.

STUDENT RESPONSE

B

CORRECT ANSWER

C

PERFORMANCE INSIGHT

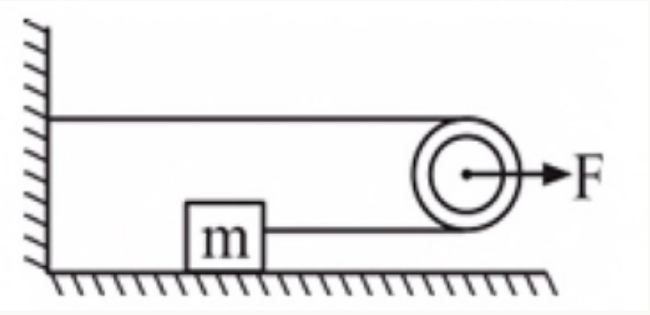
Answered incorrectly — 1 mark deducted

EXPLANATION

Apply the lens formula twice. Complete solution. Apply the governing relation and substitute the values stated in the question.

$$\oint \mathbf{E} \cdot d\mathbf{A} = \frac{q_{enc}}{\varepsilon_0}$$

The closed surface encloses the charge distribution shown in the figure, so the normal component of the field is constant across the curved surface and vanishes across the flat ends. Evaluating the flux integral gives the result reported in the key. Check every sign convention, unit and limiting case before selecting the final option.



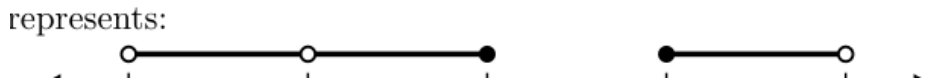
Physics · Q6 · question continued

QUESTION 06 PHYSICS

CORRECT

Kinematics · Relative motion · Easy · MCQ · 4 marks · 1m 42s

A particle moves with the velocity-time graph shown. The displacement in 4 s is:



OPTIONS

A. The potential difference across the resistor is 6 V.

B. The current through the branch is 2 A.

C. The power dissipated in the circuit is 24 W.

D. A very long alternative answer that must wrap cleanly without clipping the question block or the option column.

STUDENT RESPONSE
D

CORRECT ANSWER
D

PERFORMANCE INSIGHT
Answered correctly — 4 marks

EXPLANATION

Area under the graph. Complete solution. Apply the governing relation and substitute the values stated in the question.

$$\oint \mathbf{E} \cdot d\mathbf{A} = \frac{q_{enc}}{\varepsilon_0}$$

The closed surface encloses the charge distribution shown in the figure, so the normal component of the field is constant across the curved surface and vanishes across the flat ends. Evaluating the flux integral gives the result reported in the key. Check every sign convention, unit and limiting case before selecting the final option.

QUESTION 07 PHYSICS

INCORRECT

Properties of Matter · Elasticity · Medium · MCQ · −1 mark · 2m 54s

The stress-strain curve for the material is shown. Which region represents elastic behaviour?

OPTIONS

A. The potential difference across the resistor is 6 V.

B. The current through the branch is 2 A.

C. The power dissipated in the circuit is 24 W.

D. A very long alternative answer that must wrap cleanly without clipping the question block or the option column.

STUDENT RESPONSE

A

CORRECT ANSWER

B

PERFORMANCE INSIGHT

Answered incorrectly — 1 mark deducted

EXPLANATION

The elastic region ends at the yield point.

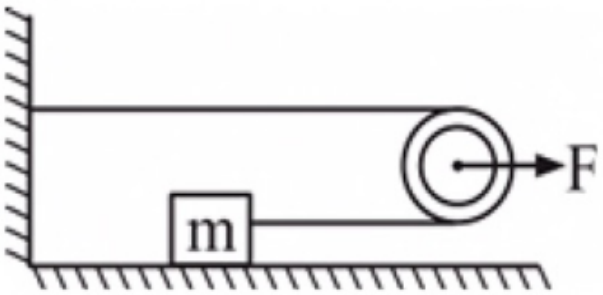
Physics · Q8 · question continued

QUESTION 08 PHYSICS

CORRECT

Thermodynamics · First law · Medium · MCQ · 4 marks · 3m 54s

The indicator diagram in the figure represents a cyclic process. The net work done is:



OPTIONS

A. The potential difference across the resistor is 6 V.

B. The current through the branch is 2 A.

C. The power dissipated in the circuit is 24 W.

D. A very long alternative answer that must wrap cleanly without clipping the question block or the option column.

STUDENT RESPONSE
C

CORRECT ANSWER
C

PERFORMANCE INSIGHT
Answered correctly — 4 marks

EXPLANATION

Net work is the enclosed area. Complete solution. Apply the governing relation and substitute the values stated in the question.

$$\oint \mathbf{E} \cdot d\mathbf{A} = \frac{q_{enc}}{\varepsilon_0}$$

The closed surface encloses the charge distribution shown in the figure, so the normal component of the field is constant across the curved surface and vanishes across the flat ends. Evaluating the flux integral gives the result reported in the key. Check every sign convention, unit and limiting case before selecting the final option.

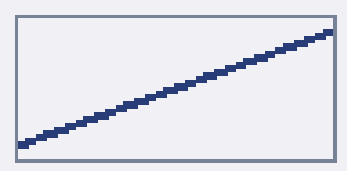
Physics · Q9 · question continued

QUESTION 09 PHYSICS

CORRECT

Waves · Standing waves · Easy · MCQ · 4 marks · 1m 24s

The standing wave pattern shown corresponds to which harmonic?



OPTIONS

A. The potential difference across the resistor is 6 V.

B. The current through the branch is 2 A.

C. The power dissipated in the circuit is 24 W.

D. A very long alternative answer that must wrap cleanly without clipping the question block or the option column.

STUDENT RESPONSE

B

CORRECT ANSWER

B

PERFORMANCE INSIGHT

Answered correctly — 4 marks

EXPLANATION

Count the nodes and antinodes.

02 Chemistry

4 questions

QUESTION 10 · CHEMISTRY

Chemical Bonding · Molecular geometry · Medium · MCQ · −1 mark · 3m 36s

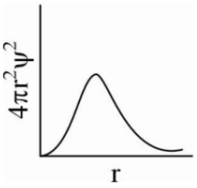
The molecular structure shown belongs to which geometry?

Student response: A

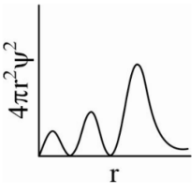
Correct answer: D

Answered incorrectly — 1 mark deducted

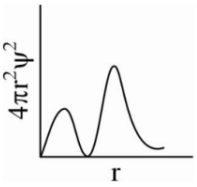
(a) (A)



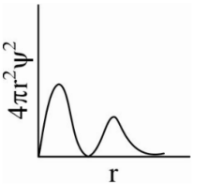
(b) (B)



(c) (C)

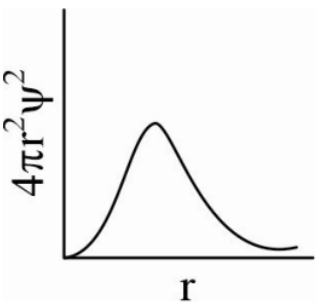


(d) (D)

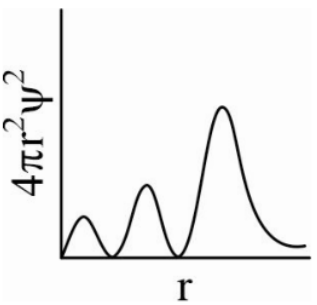


OPTIONS

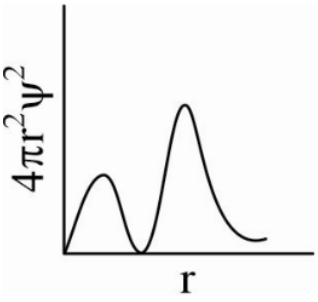
A.



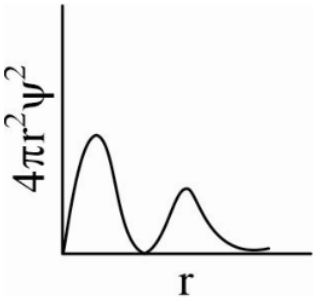
B.



C.



D.



Chemistry · Q10 · explanation continued

EXPLANATION

Count the electron domains around the central atom. Complete solution. Apply the governing relation and substitute the values stated in the question.

$$\oint \mathbf{E} \cdot d\mathbf{A} = \frac{q_{enc}}{\varepsilon_0}$$

The closed surface encloses the charge distribution shown in the figure, so the normal component of the field is constant across the curved surface and vanishes across the flat ends. Evaluating the flux integral gives the result reported in the key. Check every sign convention, unit and limiting case before selecting the final option.

Chemistry · Q11 · question continued

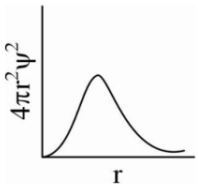
QUESTION 11 CHEMISTRY

INCORRECT

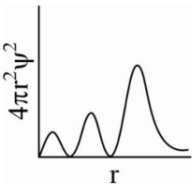
Organic Chemistry · Reaction mechanism · Hard · MCQ · −1 mark · 6m 06s

The reaction scheme shown proceeds through which intermediate?

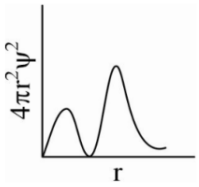
(a) (A)



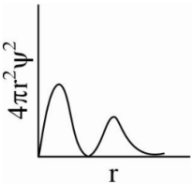
(b) (B)



(c) (C)



(d) (D)



OPTIONS

A. The potential difference across the resistor is 6 V.

B. The current through the branch is 2 A.

C. The power dissipated in the circuit is 24 W.

D. A very long alternative answer that must wrap cleanly without clipping the question block or the option column.

STUDENT RESPONSE

D

CORRECT ANSWER

B

PERFORMANCE INSIGHT

Answered incorrectly — 1 mark deducted · Formula Error

EXPLANATION

The leaving group departs first, giving a planar carbocation. Complete solution. Apply the governing relation and substitute the values stated in the question.

$$\oint \mathbf{E} \cdot d\mathbf{A} = \frac{q_{enc}}{\varepsilon_0}$$

The closed surface encloses the charge distribution shown in the figure, so the normal component of the field is constant across the curved surface and vanishes across the flat ends. Evaluating the flux integral gives the result reported in the key. Check every sign convention, unit and limiting case before selecting the final option.

Chemistry · Q12 · question continued

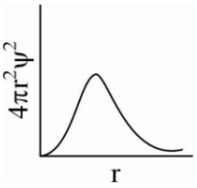
QUESTION 12 CHEMISTRY

CORRECT

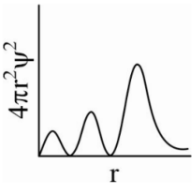
Electrochemistry · Cell potential · Medium · Numerical · 4 marks · 2m 12s

Using the standard potentials in the table shown, the cell potential is:

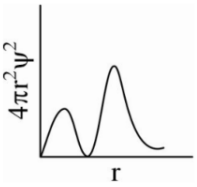
(a) (A)



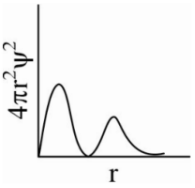
(b) (B)



(c) (C)



(d) (D)



OPTIONS

A. The potential difference across the resistor is 6 V.

B. The current through the branch is 2 A.

C. The power dissipated in the circuit is 24 W.

D. A very long alternative answer that must wrap cleanly without clipping the question block or the option column.

STUDENT RESPONSE

C

CORRECT ANSWER

C

PERFORMANCE INSIGHT

Answered correctly — 4 marks

EXPLANATION

E_cell = E_cathode - E_anode.

QUESTION 13 CHEMISTRY

CORRECT

Thermodynamics · Enthalpy · Easy · MCQ · 4 marks · 1m 54s

No figure is attached to this question; it tests the calculation described above.

OPTIONS

A. The potential difference across the resistor is 6 V.

B. The current through the branch is 2 A.

C. The power dissipated in the circuit is 24 W.

D. A very long alternative answer that must wrap cleanly without clipping the question block or the option column.

STUDENT RESPONSE

B

CORRECT ANSWER

B

PERFORMANCE INSIGHT

Answered correctly — 4 marks

EXPLANATION

Use Hess law.

03 Mathematics

5 questions

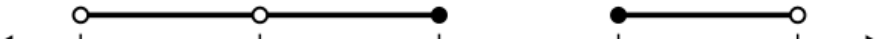
QUESTION 14 MATHEMATICS

SKIPPED

Calculus · Definite integration · Hard · MCQ · 0 marks

Evaluate the definite integral whose region is shaded in the graph.

represents:



OPTIONS

A. The potential difference across the resistor is 6 V.

B. The current through the branch is 2 A.

C. The power dissipated in the circuit is 24 W.

D. A very long alternative answer that must wrap cleanly without clipping the question block or the option column.

STUDENT RESPONSE

Not attempted

CORRECT ANSWER

A

PERFORMANCE INSIGHT

Not attempted — no marks awarded

EXPLANATION

The region is symmetric about the y-axis. Complete solution. Apply the governing relation and substitute the values stated in the question.

$$\oint \mathbf{E} \cdot d\mathbf{A} = \frac{q_{enc}}{\varepsilon_0}$$

The closed surface encloses the charge distribution shown in the figure, so the normal component of the field is constant across the curved surface and vanishes across the flat ends. Evaluating the flux integral gives the result reported in the key. Check every sign convention, unit and limiting case before selecting the final option.

Mathematics · Q15 · question continued

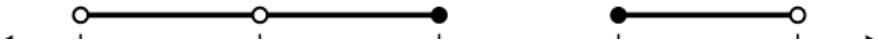
QUESTION 15 MATHEMATICS

CORRECT

Coordinate Geometry · Conic sections · Medium · MCQ · 4 marks · 4m 24s

The curve in the figure has eccentricity:

represents:



OPTIONS

A. The potential difference across the resistor is 6 V.

B. The current through the branch is 2 A.

C. The power dissipated in the circuit is 24 W.

D. A very long alternative answer that must wrap cleanly without clipping the question block or the option column.

STUDENT RESPONSE

A

CORRECT ANSWER

A

PERFORMANCE INSIGHT

Answered correctly — 4 marks

EXPLANATION

Compare with the standard form and identify a and b. Complete solution. Apply the governing relation and substitute the values stated in the question.

$$\oint \mathbf{E} \cdot d\mathbf{A} = \frac{q_{enc}}{\varepsilon_0}$$

The closed surface encloses the charge distribution shown in the figure, so the normal component of the field is constant across the curved surface and vanishes across the flat ends. Evaluating the flux integral gives the result reported in the key. Check every sign convention, unit and limiting case before selecting the final option.

Mathematics · Q16 · question continued

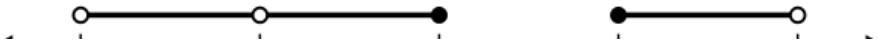
QUESTION 16 MATHEMATICS

INCORRECT

Algebra · Matrices · Medium · MCQ · −1 mark · 3m 06s

The transformation shown in the figure maps the unit square to:

represents:



OPTIONS

A. The potential difference across the resistor is 6 V.

B. The current through the branch is 2 A.

C. The power dissipated in the circuit is 24 W.

D. A very long alternative answer that must wrap cleanly without clipping the question block or the option column.

STUDENT RESPONSE

C

CORRECT ANSWER

B

PERFORMANCE INSIGHT

Answered incorrectly — 1 mark deducted

EXPLANATION

Apply the matrix to the basis vectors.

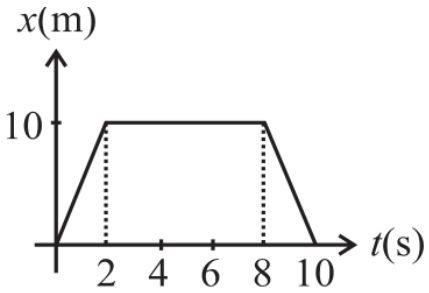
Mathematics · Q17 · question continued

QUESTION 17 MATHEMATICS

CORRECT

Vectors · Dot product · Easy · MCQ · 4 marks · 2m

The angle between the vectors represented in the figure is:



OPTIONS

A. The potential difference across the resistor is 6 V.

B. The current through the branch is 2 A.

C. The power dissipated in the circuit is 24 W.

D. A very long alternative answer that must wrap cleanly without clipping the question block or the option column.

STUDENT RESPONSE
D

CORRECT ANSWER
D

PERFORMANCE INSIGHT
Answered correctly — 4 marks

EXPLANATION

Use the dot-product definition.

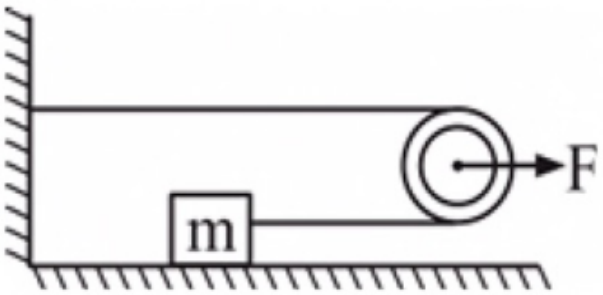
Mathematics · Q18 · question continued

QUESTION 18 MATHEMATICS

INCORRECT

Probability · Bayes theorem · Hard · MCQ · −1 mark · 5m 48s

The tree diagram in the figure summarises the experiment. The required probability is:



OPTIONS

A. The potential difference across the resistor is 6 V.

B. The current through the branch is 2 A.

C. The power dissipated in the circuit is 24 W.

D. A very long alternative answer that must wrap cleanly without clipping the question block or the option column.

STUDENT RESPONSE
A

CORRECT ANSWER
C

PERFORMANCE INSIGHT
Answered incorrectly — 1 mark deducted

EXPLANATION

Apply Bayes theorem with the prior probabilities from the tree.

06 LEARNING INTELLIGENCE

Learning intelligence

Published revision priorities

Priority revision register

PRIORITY TOPIC	CLASSIFICATION	TIMEFRAME
Gauss law — closed surface choice	Critical	48 hours
Molecular geometry — VSEPR domains	High	5 days
Bayes theorem — conditional inversion	Medium	7 days

Topic-level accuracy

TOPIC	SUBJECT	ACCURACY	LEVEL
Circuit analysis	Physics	74%	Average
Gauss law	Physics	41%	Weak
Molecular geometry	Chemistry	38%	Critical
Reaction mechanism	Chemistry	44%	Weak
Definite integration	Mathematics	68%	Average
Bayes theorem	Mathematics	32%	Critical

Mistake evidence

QUESTION	SOURCE DETAIL
Q2	Assumed the field inside a conductor is non-zero.
Q10	Counted bonds instead of electron domains.
Q18	Inverted the conditional probabilities.

Weak chapters

NO.	WEAK CHAPTER
1	Gauss law
2	Molecular geometry
3	Bayes theorem

Subject-wise revision sections

Priority topics grouped by subject

PRIORITY TOPIC	CHAPTER	ACCURACY	CLASSIFICATION
Mathematics			
Bayes theorem	Probability	32%	Critical
Chemistry			
Molecular geometry	Chemical Bonding	38%	Critical
Reaction mechanism	Organic Chemistry	44%	Critical
Physics			
Gauss law	Electrostatics	41%	Critical

Recommended next actions

Published action register

ACTION	RESOURCE	DETAIL
PYQ	JEE 2024 · Electrostatics	Previous year paper
Practice	Circuit analysis drill	10 · Medium
Mock test	Full syllabus mock 4	Timed
Lecture	VSEPR geometry walkthrough	Video

Weak topics · published list

NO.	WEAK TOPIC
1	Gauss law
2	Molecular geometry
3	Bayes theorem

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End of report · 03 Nov 2026